



RAMCO INSTITUTE OF TECHNOLOGY

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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electronics and Communication Engineering

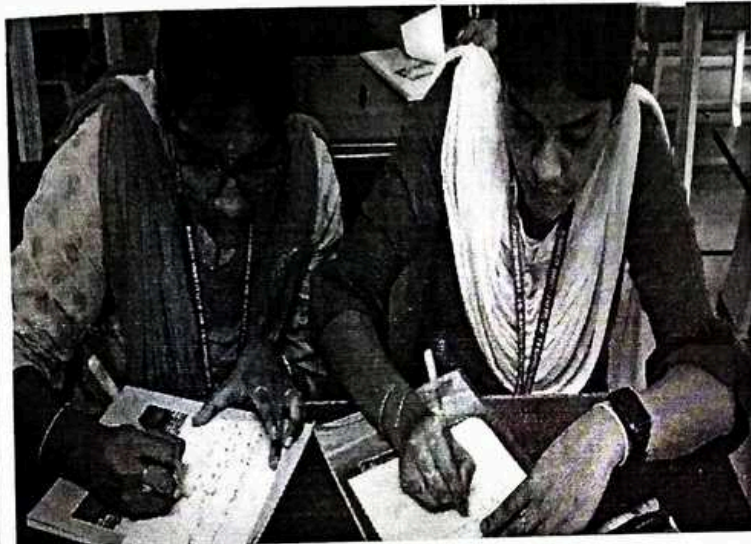
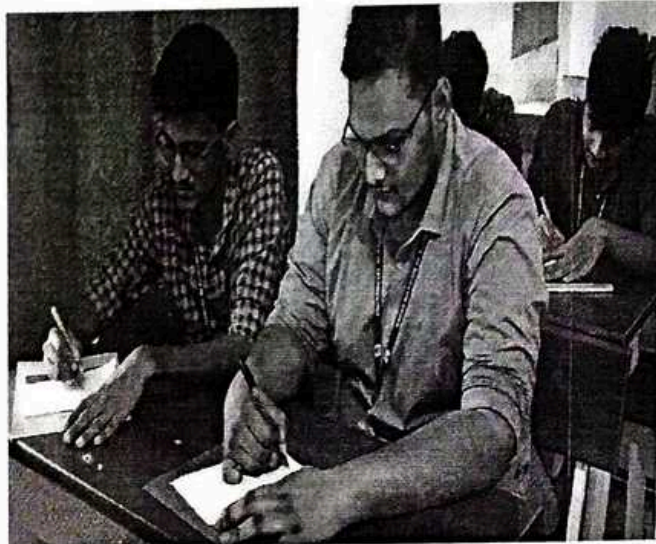
Academic Year: 2022 – 2023(ODD Semester)

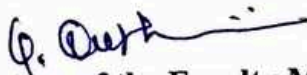
INNOVATIVE TEACHING METHODS

Degree, Semester & Branch : VII Semester B.E. ECE

Course Code & Title : EC8791 & Embedded and Real Time Systems

Name of the Faculty member: Mrs.G.Subhashini

SL.N o.	Date	Topic(s)	Activity*	Reference
UNIT I - Introduction to Embedded System Design				
1.	23.08.2022	Designing with Computing Platforms	One Minute Paper	https://oncourseworkshop.com/self-awareness/one-minute-paper/
				


Signature of the Faculty Member


24/09/22
HOD



Department of Electronics and Communication Engineering
Academic Year 2022-2023 (ODD Semester)
Planning Document

Degree, Semester & Branch: VII Semester B.E. ECE

Course Code & Title: EC8791 & Embedded and Real Time Systems

Name of the Faculty member: Mrs.G.Subhashini, AP/ECE

- **Unit/Topic:** Unit-II/ Instruction Set
- **Course Outcome:** CO2
- **Topic Learning Outcome:** TLO2.1
- **Activity Chosen:** Jigsaw

Learning Outcomes:

The Student will be able to

- ❖ Understand the techniques used in Instruction Set.
- ❖ Know the concept of Instruction Set
- ❖ Participate actively in the learning process of Instruction Set
- ❖ Discuss the concepts, convey idea and share the views of Instruction Set and its types.
- ❖ Communicate effectively by sharing their views among the teammates.

Justification:

The basic concept of Instruction Set can be explained with Jigsaw method. The Jigsaw Strategy is an efficient way to learn the course material in a cooperative learning style. The jigsaw process encourages listening, engagement, and empathy by giving each member of the group an essential part to play in the academic activity. The activity 'JIGSAW' is used to recall the concepts learnt and enhance the learning through this collaboration

- **Time Allotted for the Activity:** 50 Minutes
- **Details of the Implementation:**

(i) Materials for the Activity:

The materials for the preparation of the students will be shared one week before through LMS Canvas. Including the material, students were asked to refer to the web content also.

References:

Book Title/ Author/ Publisher/ Edition	Page No.
1. Marilyn Wolf, "Computers as Components - Principles of Embedded Computing System Design", Third Edition "Morgan Kaufmann Publisher (An imprint from Elsevier), 2012.	840-843
Websites	

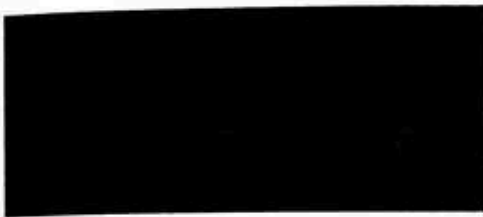
- The Time management will be avoided by using timer clock for 10 minutes duration.

JIGSAW

Round 1: Focus Groups

- Divide students into group and give each group a different problem to solve

1	2	3
4	5	6



1	2	3
4	5	6

1	2	3
4	5	6



JIGSAW

Round 2: Task Groups

- Mix the groups so that students can bring their specific focus to common problem

1		1
1	1	

2		2
2	2	

3		3
3	3	

4		4
4	4	

5		5
5	5	

6		6
6	6	

GROUP 1

Topic: Instruction Set

1		
1	1	

GROUP 2

Topic: Reduced Instruction Set Computer

2		
2	2	

GROUP 3

Topic: Complex Instruction Set Computer

3		
3	3	

GROUP 4

Topic: Why Instruction Sets are important

4		
4	4	

GROUP 5

Topic: Enhanced Instruction set

5		
5	5	

GROUP 6

Topic: How the Instruction Commands used

6		
6	6	

• **Student Feedback: Written Feedback**

1. How do you rate the activity for learning the concept? Give tick mark (✓)			
Excellent	Good	Satisfactory	Poor
2. How do you rate the reference material provided related to the topic? Give tick mark (✓)			
Excellent	Good	Satisfactory	Poor
3. Have you enjoyed the activity? Give tick mark (✓)			Yes: No:
4. Challenges faced during activity/presentation. Give tick mark (✓)			Yes: No:
If yes what?:			
Name & Signature of the Student with date			


Signature of Faculty Member


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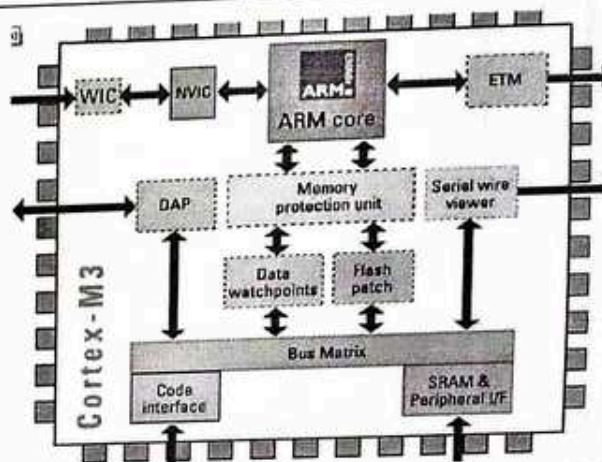
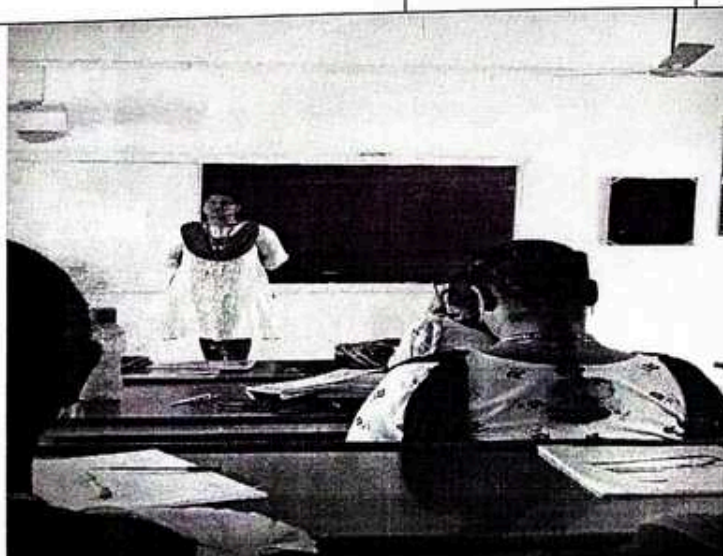
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Academic Year: 2022 – 2023(ODD Semester)

INNOVATIVE TEACHING METHODS

Degree, Semester & Branch : VII Semester B.E. ECE B
Course Code & Title : EC8791 & Embedded and Real Time Systems
Name of the Faculty member: Mrs.G.Subhashini

Name of the Faculty		SL.N o.	Date	Topic(s)	Activity*	Reference
UNIT II - Introduction to Embedded System Design						
1.	13.9.2022	ARM Cortex M3 MCU		Mind Map	https://www.mindmeister.com/blog/students-guide-to-mind-mapping/	
						
UNIT III - Embedded Programming						
2	20.09.2022	Models of Programmes		Role Play	https://www.niu.edu/citl/resources/guides/instructional-guide/role-playing.shtml	
						

G. Subhashini
15/11/22

Signature of the Faculty Member

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15/11/22
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Academic Year 2022-2023 (ODD Semester)

Planning Document

Degree, Semester & Branch: VII Semester B.E. ECE B

Course Code & Title: EC8791 & Embedded and Real Time Systems

Name of the Faculty member: Mrs.G.Subhashini, AP/ECE

- **Unit/Topic:** Unit-V / : Task Assignment
- **Course Outcome:** CO4
- **Topic Learning Outcome:** TLO 4.2
- **Activity Chosen:** Flipped Classroom

Learning Objective and Outcomes:

Objective: To explain the working principle of task assignment with various functions like Static, Dynamic, Focussed Addressing and Binding (FAB), The Buddy Strategy Algorithm and Utilization-Balancing Algorithm

Outcome:

At the end of the flipped class activity,

- ❖ The students will be able to explain the various functionality of task assignment with real time examples in the areas of Embedded Systems and Microprocessor and Micro controllers.
- ❖ The self-learning skill and team work skills will be enhanced.

Justification for Choosing the Topic using Flipped Class:

The topic was chosen for flipped class as there are many applications for Embedded Systems. The Unit 4 deals with the details of Structure of Real Time Systems, Estimating Program Run Times, Task Assignment and Task Scheduling. Since the students are exposed to the details of processor, I thought of giving the topic '*Task Assignment*' for flipped class. This topic can be learnt in self-learning mode.

As there are many functions of Scheduling only some of the functions were discussed in their prescribed text book. If this flipped class is used for the topic apart from the learning materials given by the faculty, students have chance to browse and find more information related to functionality of task Assignment. Thus the task assignment with various functions

like Static, Dynamic, Focussed Addressing and Binding (FAB), The Buddy Strategy Algorithm and Utilization-Balancing Algorithm will be covered in 2 periods through this activity which is not possible in traditional teaching. This is the reason for choosing this topic for flipped class activity.

Learning Material:

The learning material for applications of digital signal processors will be book chapter and video lectures

1. Book Chapter:

Book Title	-	Embedded electronics
Author	-	Brett D DePaola
Publisher	-	Publishing Ltd 2021
Edition	-	Second Edition,

2. Book Chapter:

Book Title	-	The Embedded Rust Boo
Author	-	M.J. Smith
Publisher	-	Addisson Wesley

3. MIT Open courseware:

Title	-	Scheduling
Professor	-	Prof. Lawrence Sass
Topics covered	-	Different Rask Assignment
Web link	-	https://ocw.mit.edu/courses/4-510-digital-design-fabrication-fall-2008/pages/syllabus/
Time Duration	-	17 Minutes

4. NPTEL Video:

Title	-	<u>Scheduling Policies for a Real Time Operating System</u>
Lecture series	-	21
Course name	-	Embedded Systems
Professor	-	Dr.Santanu Chaudhury
Web link	-	http://www.infocobuild.com/education/audio-video-courses/electronics/EmbeddedSystems-IIT-Delhi/lecture-12.html
Time Duration	-	55 Minutes

The book chapters will be shared to the students as materials and video like from NPTEL and MIT open courseware will be posted related Task Assignment in the course website.

The learning material of book chapters consist of details of Task Assignment as

1. Task Assignment
2. 2. Scheduling'
3. Static Scheduling
4. Dynamic Scheduling
5. Utilization Balancing Algorithm
6. Next-Fit Algorithm for RMA
7. Bin Packing Algorithm for EDF

The NPTEL video discusses about Tin detail and MIT open sk Assignmentcourseware video shows the Architecture and application with demonstration of Embedded Systems.

Creating a script:

The script will be *created with the audio of 2 minutes* for the instructions to the students about the reasons and portions for special attention.

Script:

Dear Students,

I hope you will have a new experience of flipped class for the topic Task Assignment as discussed already. The objective of the activity is flip of traditional classroom where learning will happen outside the class and working, homework's etc will happen in the class. As narrated earlier, the learning materials are posted in the course website and the portions to be concentrated will preparing the presentations are

The book chapter learning material for the applications of digital signal processors consist of all the details of each application. In which the first book chapter and second book chapters covers the working of Task Assignment.

The **MIT open courseware: Video lecture of 17 minutes** gives the detail demonstration of real time application which gives an idea of how the DSP processors are used.

The **NPTEL video** gives the detailed presentation of Task Assignment if required to know about the peripherals used for certain application are dealt in detail. Kindly make use of book chapters for preparing the presentations and apart from the details any other information regarding the working principle can also be added from other online resources with proper citations.

Prepare a presentation for 10 minutes containing the

- ✓ Introduction
- ✓ How Task Assignment play a role in the application related to your group
- ✓ Block diagram
- ✓ Technical Information's
- ✓ Design Flow

- ✓ Testing Methods
- ✓ Advantages and Disadvantages

Submit the handouts of the presentation on the day of activity. Hope you will enjoy the flipped class!

Thank you students.

Formation of the groups and Justification:

The formation of group for flipped class will be given to student's choice.

Number of groups: 8

Class Strength: 41

Members per group: 5 (7 groups) and 6 (1 group)

The choice of the group type is heterogeneous grouping, the condition of the grouping will be informed to the students. i.e. A group should be a mixer of advance learners, average learners and slow learners. The students group was given the students' choices with the above said conditions because the students should learn about Task Assignment and prepare a presentation about the topic. Since a detailed presentation should be made by each group, if the choice is given to the student's only successful learning will happen. The list of group is shown in the evaluation table for flipped class given below.

Plan for implementing this activity:

Outside class activity:

- ✓ Reading of learning material: 10 days time duration
- ✓ Preparation of presentation: 4 days time duration
- ✓ Discussions and clarification of doubts : Any time within the 2 weeks time for discussion within the group mates and any clarification regarding the flipped class

Inside class activity:

- ✓ Presenting the topic assigned by each group – 10 minutes duration
- ✓ Question & Answering, oral feedback – 3 minutes duration

Implementation process:

- ❖ The students were informed about the flipped class activity at least 2 weeks before the implementation of the activity.
- ❖ The materials like documents, videos related to the topic will be posted in the course website - Canvas.

- ❖ The grouping of students will also be done well in advance and each group is allotted with one topic of Task Assignment of their choice among the sub topic of Task Assignment listed. (2 weeks before implementation)
- ❖ An announcement will be posted once in two days about the activity as reminder.
- ❖ A **Discussion forum** will be opened for flipped class, in which the group members can use for discussion outside class. Using the '*your questions about flipped class*' – doubts of the students can be posted and the instructor will clarify within the 8 hours of post.
- ❖ A **whatsapp group** will be created for flipped class activity to clarify the doubts and discussions.
- ❖ The students have to prepare a presentation for 10 minutes with the concept learnt and 3 minutes for Question & answering and tell about the contribution of each member.
- ❖ The inside class activity: presentation will be planned for 2 periods. In which each group has to present their topic chosen and others will also listen to the presentation.
- ❖ The basic principles of Task Assignment Building block architecture are covered to the whole class and the handouts prepared on the topic will be shared within the group members.
- ❖ Each group will be assessed through the content of presentation, communication skills, Q& A, contribution by each member as shown in the table below. Both formative and summative assessment will be made.
 - ✓ Content of presentation (20 Marks)
 - ✓ Communication skills (10 Marks)
 - ✓ Q & A(10 Marks)
 - ✓ Contribution (10 Marks)
- ❖ The main objective of the flipped class activity is each group will learn one function of Task Assignment presentation helps the other members also to know about all the pre layout and post layout functionalities.
- Thus the working of task assignment with various functions like Static, Dynamic, Focussed Addressing and Binding (FAB), The Buddy Strategy Algorithm and Utilization-Balancing Algorithm will be covered in 2 periods through this activity which is not possible in traditional teaching
- ❖ After completion of presentation, both oral and written feedbacks will be collected about the flipped class from the students.
- ❖ The evaluation on the flipped class is done finally and will be assessed based on the following parameters. The marks will be considered for final internal mark calculation.

Expected Difficulties:

- ❖ **Lack of preparation:** Making the students to learn the materials is the challenging task. Each student has to read the material posted in the course website.
- ❖ Non participation in the activity: All the students should be made involve in the activity
- ❖ Time management: The flipped class activity should be able to complete in the planned duration.

Plan for preventing these difficulties:

- ❖ Making each student accountable by making formative and summative assessment for individual and group performance.
- ❖ Including the assessment mark for final internal mark calculation makes each student to participate actively in the flipped class activity.
- ❖ Each student contribution mark in the assessment will be based on the discussions in the course website and whatsapp group through which all the students will participate in the activity.
- ❖ Posting announcements periodically for learning and preparing the presentation.
- ❖ The Time management will be avoided by using timer clock for 10 minutes duration.

Hope by conducting this flipped class the students will have different experience compared to traditional class teaching. This activity helps them to enhance the self learning skills and team work.

Student Feedback to access the success of the flipped class:

The feedback about the success of the flipped class will be collected in two modes

- ❖ **Oral feedback**
- ❖ **Written Feedback**

Oral Feedback:

The oral feedback will be collected at the end of the presentation by the students about

- ✓ Experience in doing this flipped class
- ✓ Contribution by each member of the group.
- ✓ What was the learning outcome of such activity?
- ✓ What is the difference you felt between traditional teaching and flipped class?

Written Feedback:

A written feedback will be collected with the following questionnaire at the end of completing the activity.

Students Feedback: Flipped Class Room Activity

1. How do you rate the activity for learning the concept of Application Examples of DSP Processors?

☐ Excellent ☐ Good ☐ satisfactory ☐ Poor

2. How do you rate the reference material provided related to the topic?

☐ Excellent ☐ Good ☐ satisfactory ☐ Poor

3. Share your experience in doing this activity

4. Have you enjoyed the activity? Yes/ No
5. State 2 benefits and 2 challenge you faced during preparation.
- Benefits: _____

Challenges: _____

Name & Signature of the Student with date

Based on the feedback collected, the success of the activity can be assessed and from the challenges faced during preparation the problems can be identified and rectified in the future flipped class activities.


11/10/22
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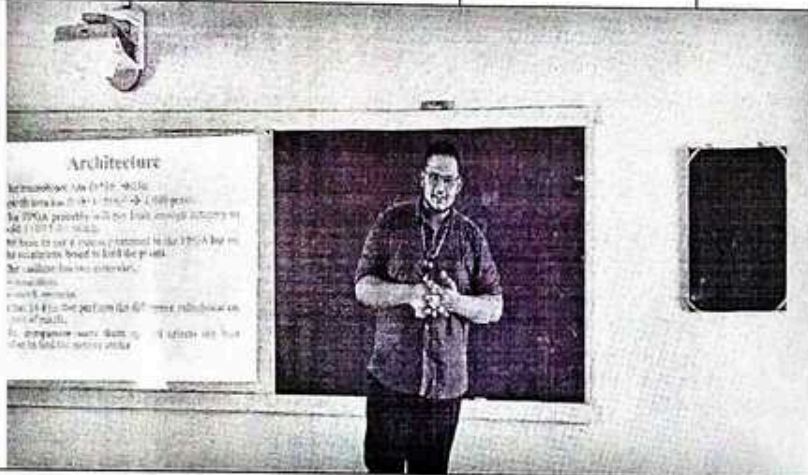

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Department of Electronics and Communication Engineering
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Name of the Faculty member: Mrs.G.Subhashini

Sl.No.	Date	Topic(s)	Activity*	Reference
UNIT IV – Real-Time Systems				
1	15.10.2022	Reliability	Head and Talk	https://gdhr.wa.gov.au/learning/teaching-strategies/finding-out/head-talk
				
UNIT V - Processes and Operating Systems				
2.	03.11.2022	Distributed Embedded Systems	Open Book Test	https://www.educationcorner.com/openbook-tests.html/
				

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